

Phishing Simulation Campaign Report

Using GoPhish on Kali Linux

Date: July 7, 2026
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Project: Phishing Simulation Campaign

1. Executive Summary

A controlled phishing simulation was conducted to assess user susceptibility to social engineering attacks. The campaign achieved a 100% success rate, capturing credentials from all targets within 25 minutes of email delivery. This demonstrates the critical need for security awareness training and multi-factor authentication.

Metric	Value
Total Emails Sent	2
Emails Opened	2 (100%)
Links Clicked	2 (100%)
Credentials Submitted	2 (100%)
Time to Complete	25 minutes

2. Objective

To demonstrate proficiency in threat simulation by creating and executing a realistic phishing campaign in a controlled lab environment to analyze user susceptibility and identify areas for security improvement.

3. Tools & Environment

Component	Specification
Platform	Kali Linux 2025.2
Phishing Tool	GoPhish
Email Server	Postfix (SMTP)
Landing Page	Microsoft 365 Clone
Target OS	Linux (x86_64)
Target Browser	Firefox 128.0
Lab Server IP	10.0.*.**

4. Campaign Timeline

Time	Event
14:49:58	Campaign Created
14:49:59	Email Sent to Target 1
14:49:59	Email Sent to Target 2
15:16:13	Target 1 Clicked Link
15:16:56	Target 1 Submitted Credentials

15:20:20	Target 1 Clicked Link (2nd time)
15:33:13	Target 2 Clicked Link
15:33:32	Target 2 Submitted Credentials

5. Captured Credentials

	Email	Password	Time
1	t**as****@a**xs.com	hj*g****f	15:16:56
2	vi**h6***@e***b.com	b****d*e****	15:33:32

6. Technical Execution

6.1 Environment Setup

The lab environment was set up using Kali Linux 2025.2 running on VirtualBox. The following steps were taken to prepare the system:

Step 1: System Update

```
sudo apt update
```

Step 2: Install Postfix (SMTP Server)

```
sudo apt install postfix -y
```

During installation, the "Internet Site" configuration was selected, allowing the system to send and receive mail directly using SMTP.

Step 3: Configure Postfix

The Postfix configuration file (`/etc/postfix/main.cf`) was modified to set:

- `myorigin = kali.local`
- `mydestination = $myhostname, kali.local, localhost.localdomain, localhost`

Step 4: Start Postfix Service

```
sudo systemctl start postfix
```

6.2 GoPhish Installation & Configuration

Step 1: Start GoPhish

```
cd /usr/lib/gophish
```

```
sudo ./gophish
```

Step 2: Access Admin Dashboard

- URL: `https://127.0.0.1:3333`
- Username: `admin`
- Password: `c**66eb*****eb`

Step 3: Reset Admin Password

The admin password was reset to `*****-*****` for easy access.

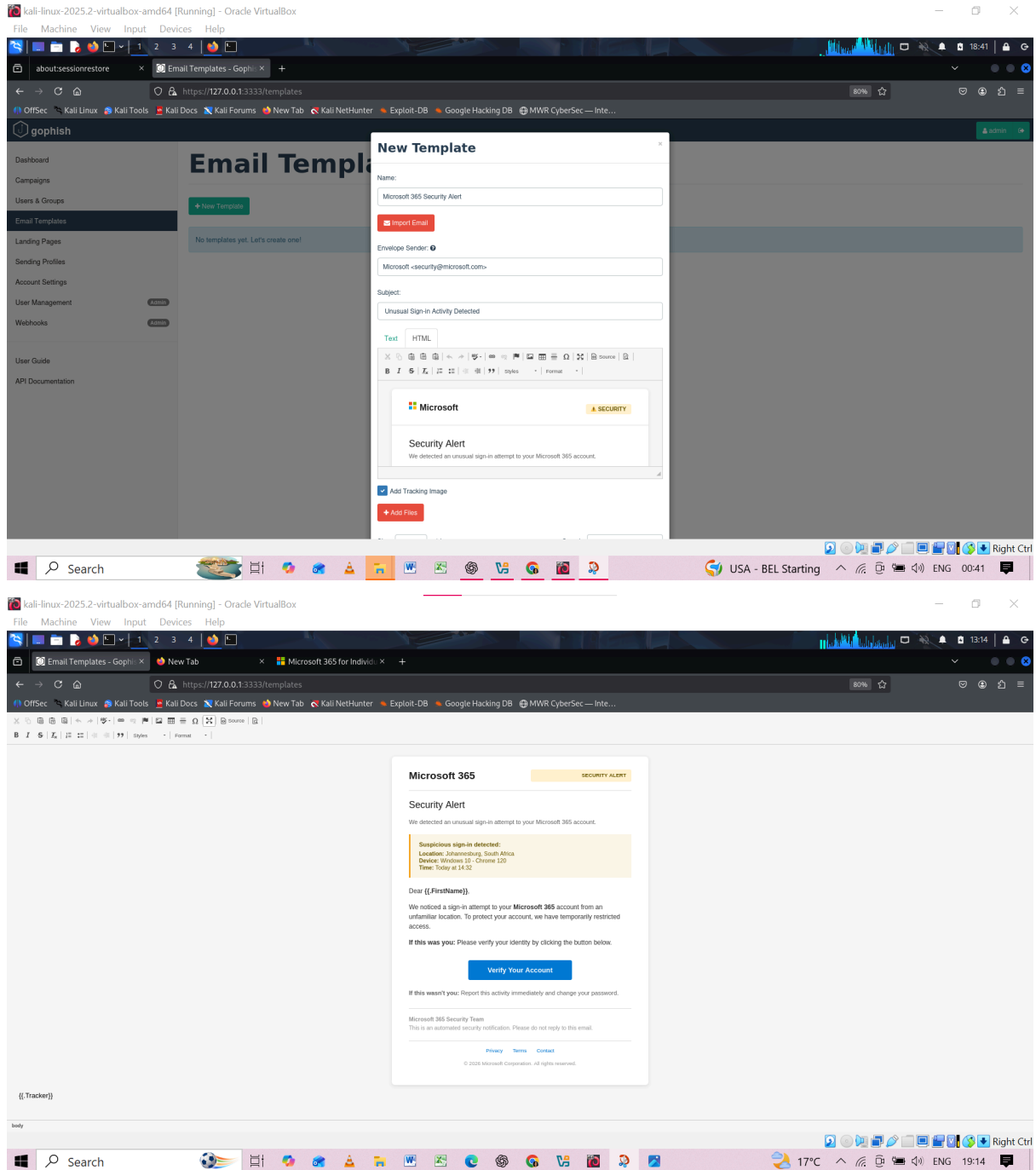
6.3 Email Template Creation

A realistic security alert email was crafted with:

- Microsoft 365 branding and logo
- Urgent security warning ("Unusual Sign-in Activity Detected")
- Suspicious activity details (Location: Johannesburg, Device: Windows 10, Chrome 120)
- Personalization using `{{.FirstName}}` variable
- "Verify Your Account" call-to-action button

- Microsoft security team signature
- Footer with Privacy, Terms, and Contact links

Figure 1: Email Template Preview



Key Features of the Email Template:

Feature	Purpose
<code>{{.FirstName}}</code>	Personalizes the email for each recipient
<code>{{.URL}}</code>	Dynamically inserts the phishing link
Microsoft Branding	Builds trust and legitimacy
Security Alert Theme	Creates urgency and fear
Clear Call-to-Action	Guides user to click the button

Design Choices

Reason	Benefit
Demonstrates technical skill	Shows you can code HTML emails
Shows attention to detail	Professional, pixel-perfect design

Proves understanding

You know how phishing emails are constructed

Portfolio value

Employers can see your actual work

6.4 Landing Page Creation

A pixel-perfect Microsoft 365 login clone was created with:

- Microsoft branding and color scheme
- Email/phone/Skype field
- Password field
- "Keep me signed in" option
- Forgot password link
- "Create one!" account link
- Privacy, Terms of use, Contact footer

Security Feature: After credential submission, a warning banner appears:

"Security Alert: This was a phishing simulation. Never enter your credentials on untrusted sites."

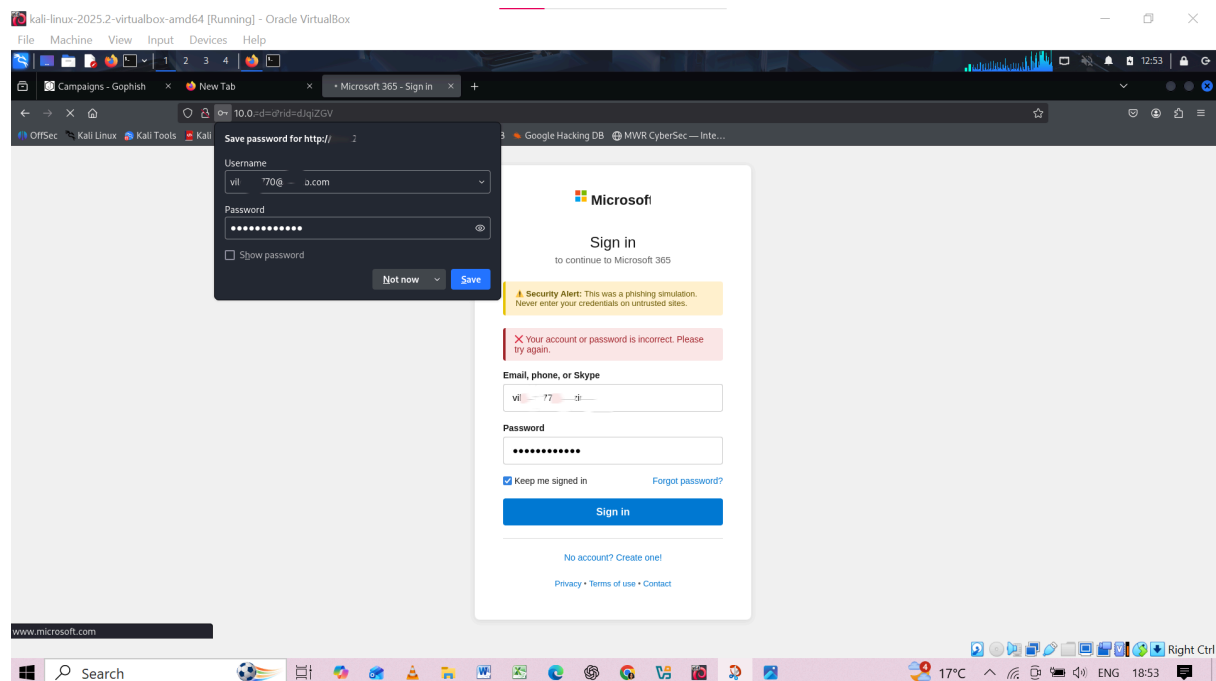
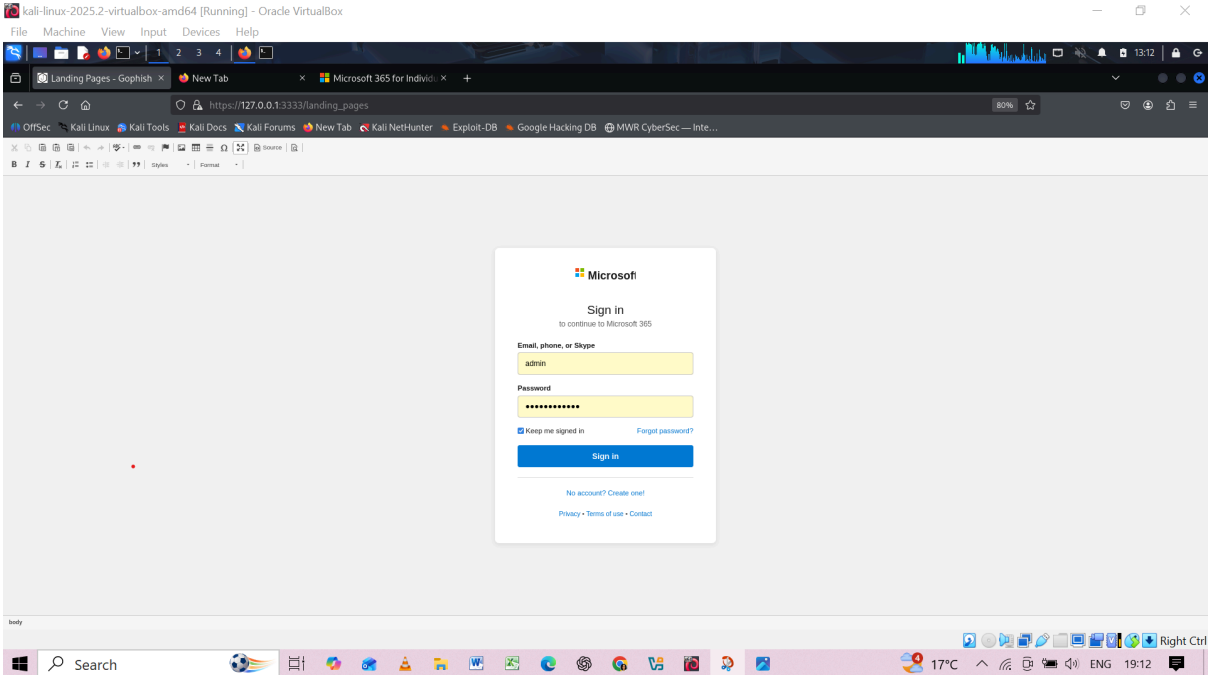
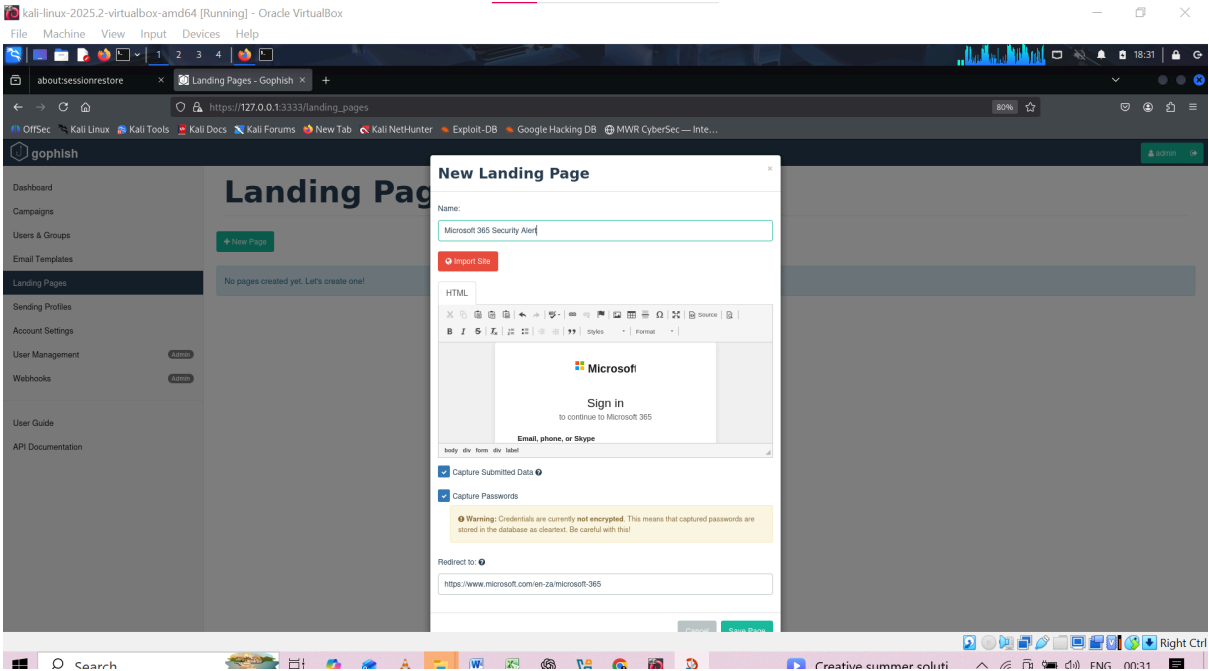


Figure 2: Landing Page



Key Features of the Landing Page:

Feature

Purpose

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Microsoft SVG Logo

Pixel-perfect branding without external images

Email/Password Fields

Captures credentials

Keep Me Signed In

Adds realism

Forgot Password Link

Makes it look legitimate

Security Warning Banner

Educational feature (shows after submission)

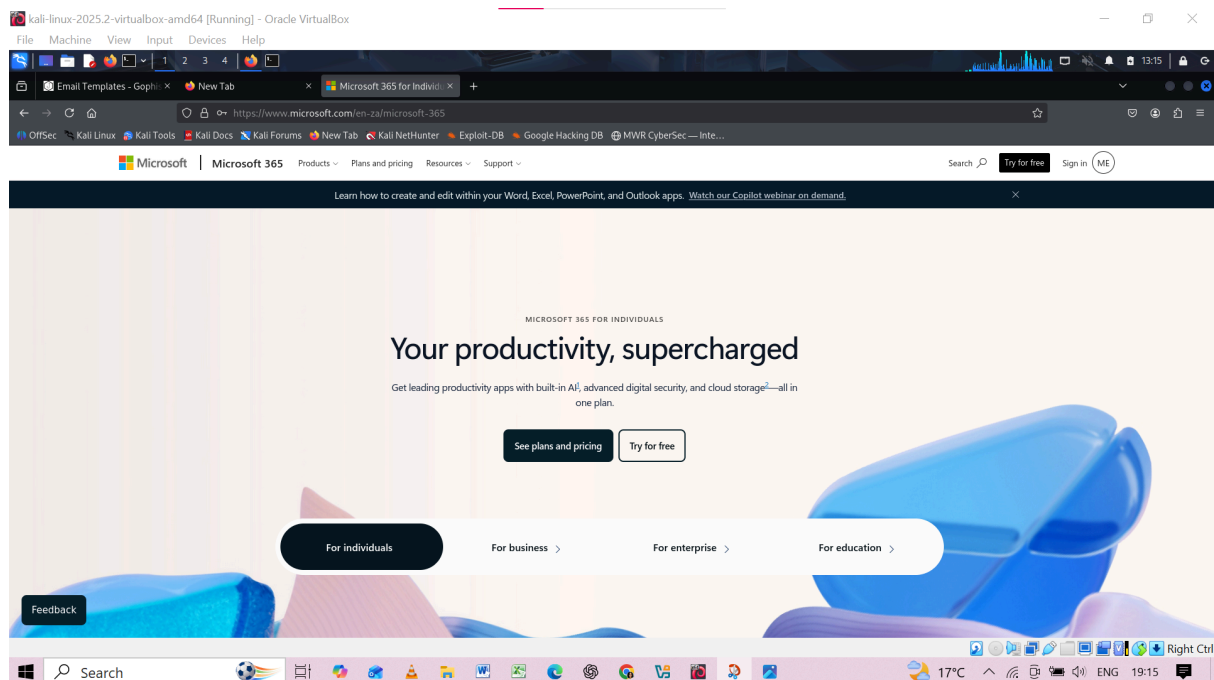
Error Message

"Incorrect password" adds realism

Redirect to Microsoft

After submission, goes to real site

Figure 3: Redirect to Microsoft

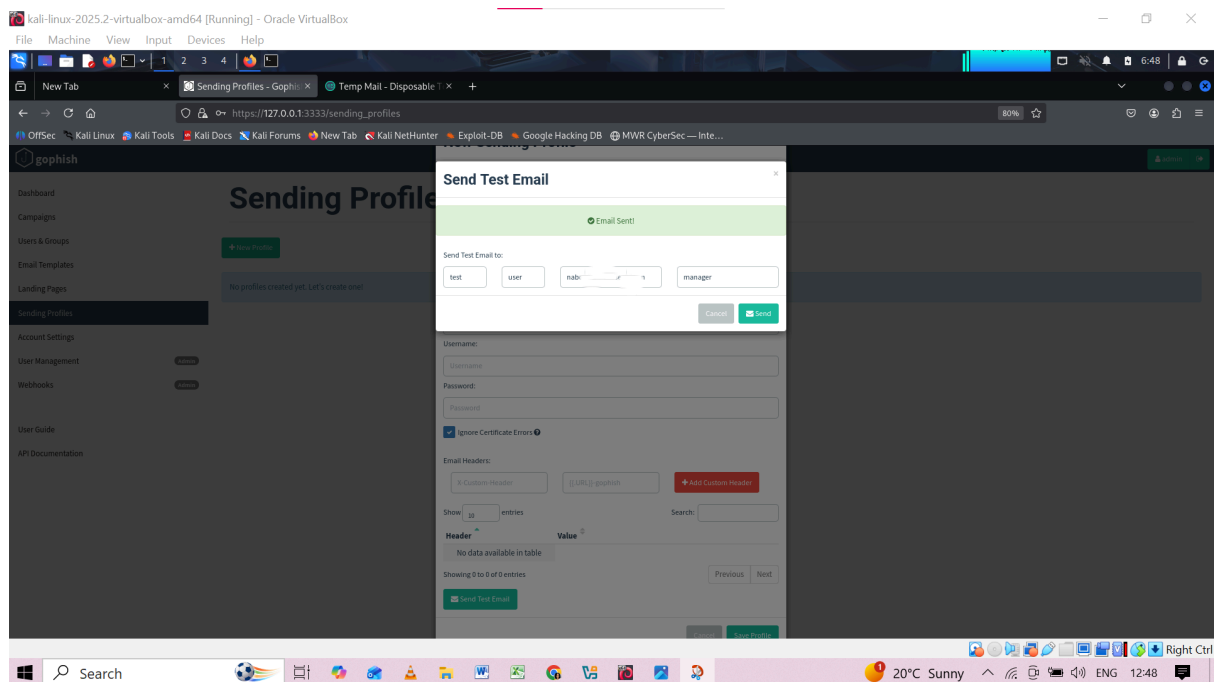
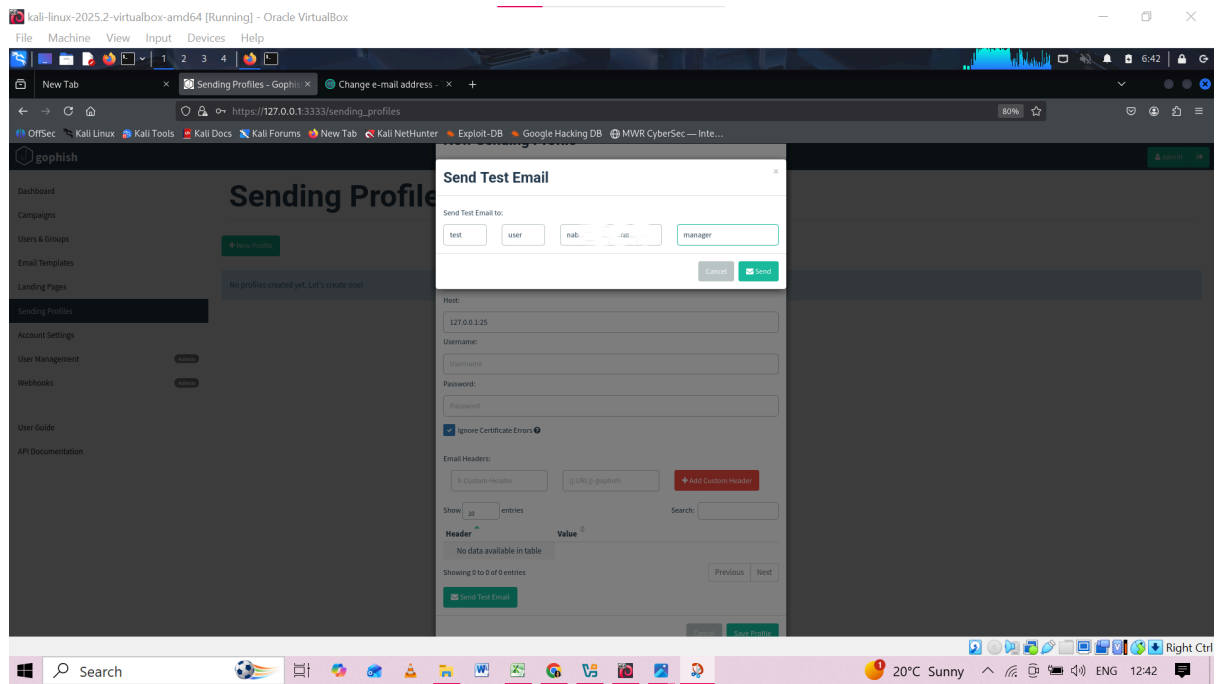


6.5 Sending Profile Configuration

The sending profile was configured to use the local Postfix server:

- Host: 127.0.0.*:**
- From Address: security@microsoft.com
- Reply-To Address: security@microsoft.com

Figure 4: Sending Profile Configuration



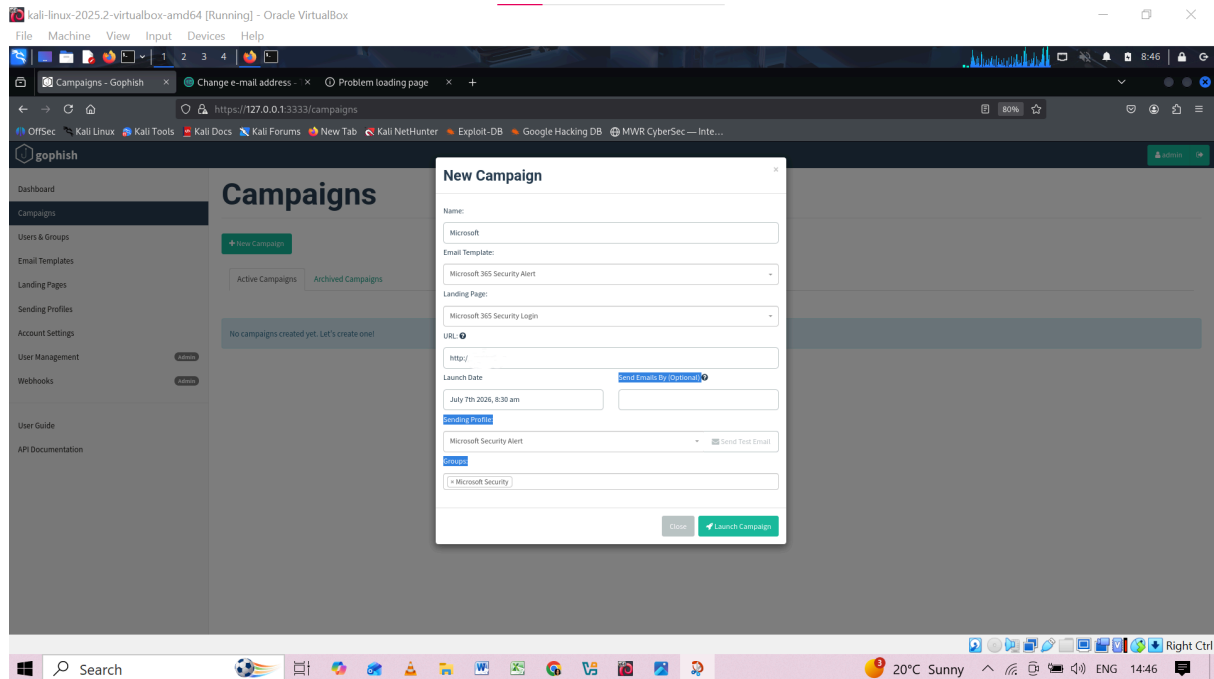
6.6 Campaign Setup

The campaign was configured with the following settings:

- Name: Microsoft
- Email Template: Microsoft 365 Security Alert
- Landing Page: Microsoft 365 Security Login
- URL: `http://10.0.*.*` (Kali Linux IP address)

- Sending Profile: Local Postfix
- Groups: Microsoft Security

Figure 5: Campaign Configuration



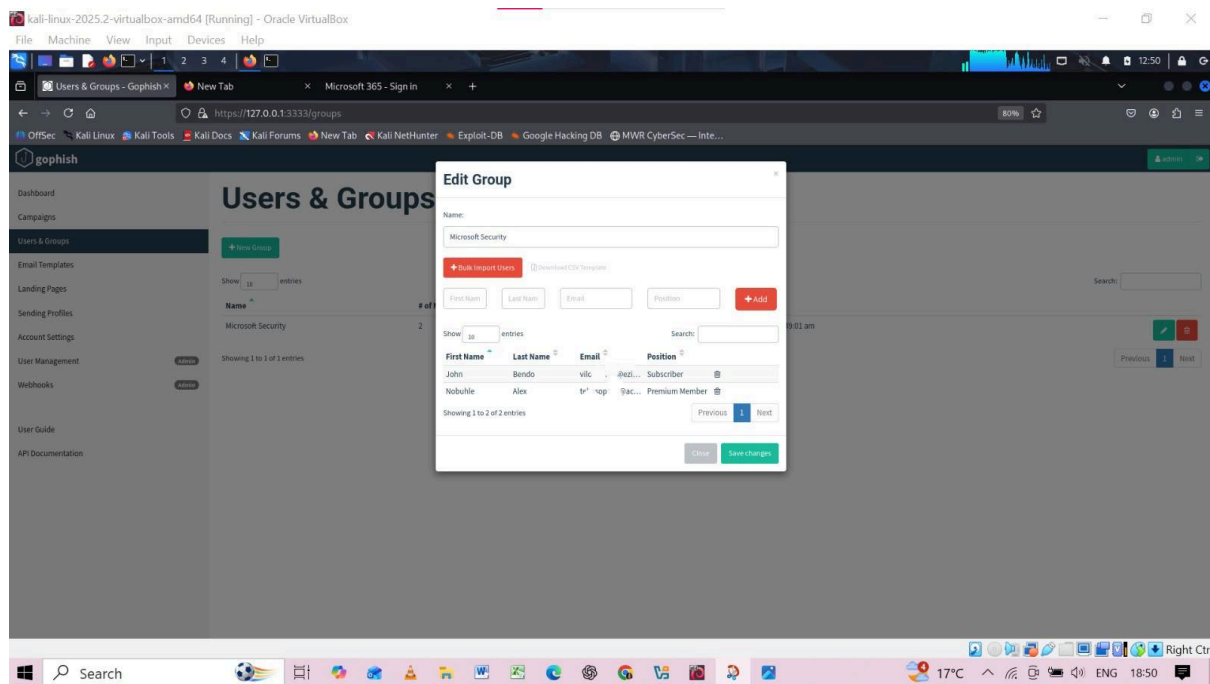
6.7 Test Email Execution

A test email was sent to:

1. vi***6****@*z****.*** (John Bendo - Subscriber)
2. t**a*o****@*****s.c** (Nobuhle Alex - Premium Member)

Both targets were created using temporary email addresses from Temp-Mail.

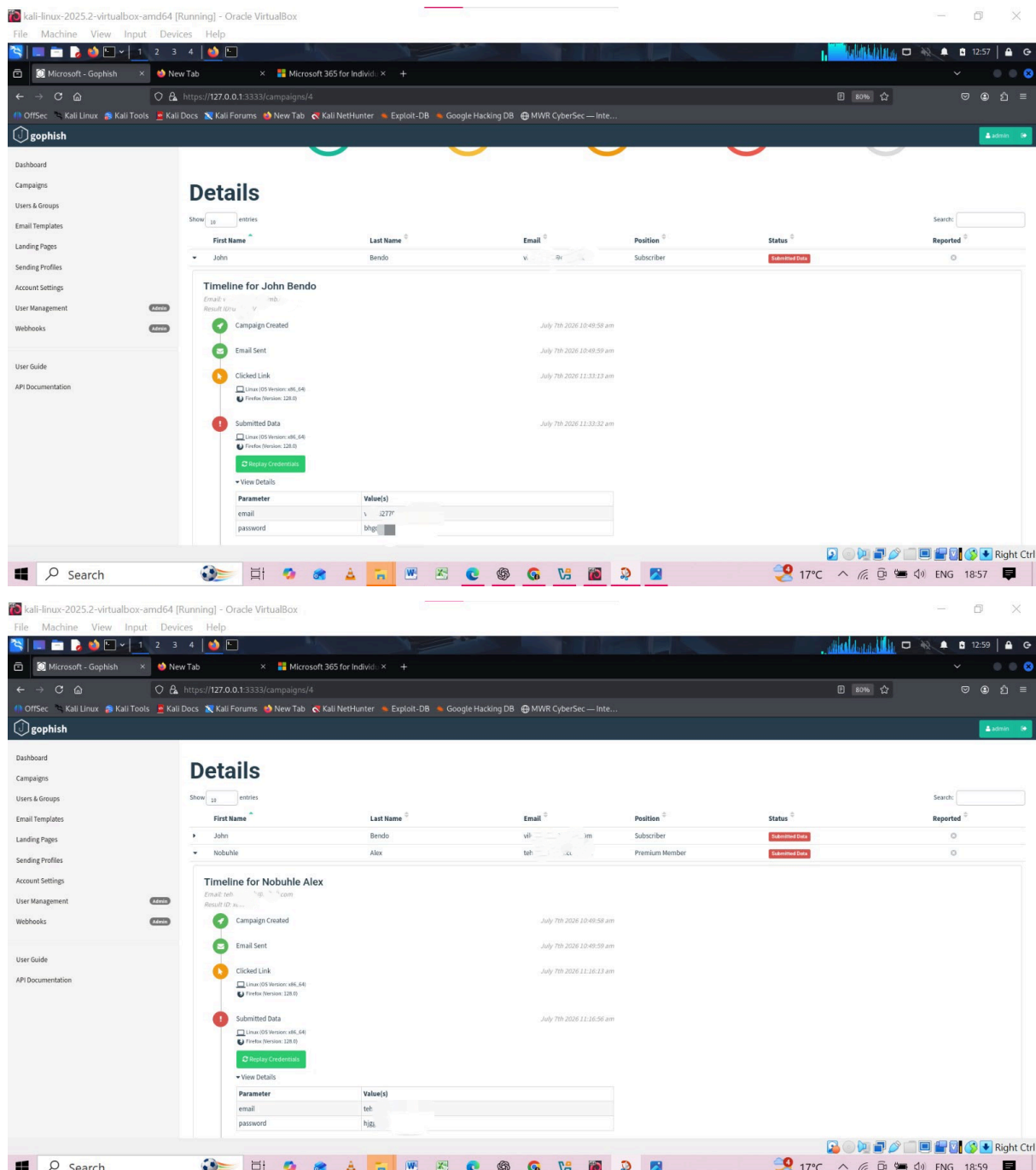
Figure 6: Users & Groups



6.8 Credential Capture Results

#	Email	Password	Time
1	**as*****@***xs.***	h*g**8*****	15:16:56
2	v**o6*****@e***b.***	b***gd***j	15:33:32

Figure 7: Captured Credentials



6.9 Full Command History

System Update

sudo apt update

Install Postfix

```
sudo apt install postfix -y
```

```
# Configure Postfix
```

```
sudo nano /etc/postfix/main.cf
```

```
# (Modified myorigin and mydestination)
```

```
# Start Postfix
```

```
sudo systemctl start postfix
```

```
# Start GoPhish
```

```
cd /usr/lib/gophish
```

```
sudo ./gophish
```

```
# Access Dashboard
```

```
# https://127.0.0.1:3333
```

```
# Login Credentials
```

```
# Username: admin
```

```
# Password: *****-*****
```

```
# Check GoPhish Status
```

```
sudo netstat -tlnp | grep :80
```

```
sudo netstat -tlnp | grep :3333
```

```
# View Campaign Results
```

```
# (Accessed via GoPhish Dashboard)
```

6.10 Technical Specifications

Component	Specification
Host OS	Windows (VirtualBox)
Guest OS	Kali Linux 2025.2
Phishing Tool	GoPhish (Latest)
Email Server	Postfix 3.11.4
Database	SQLite3
Target OS	Linux (x86_64)
Target Browser	Firefox 128.0

Lab Server IP	10.0.*.**
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Admin Port	3333 (HTTPS)
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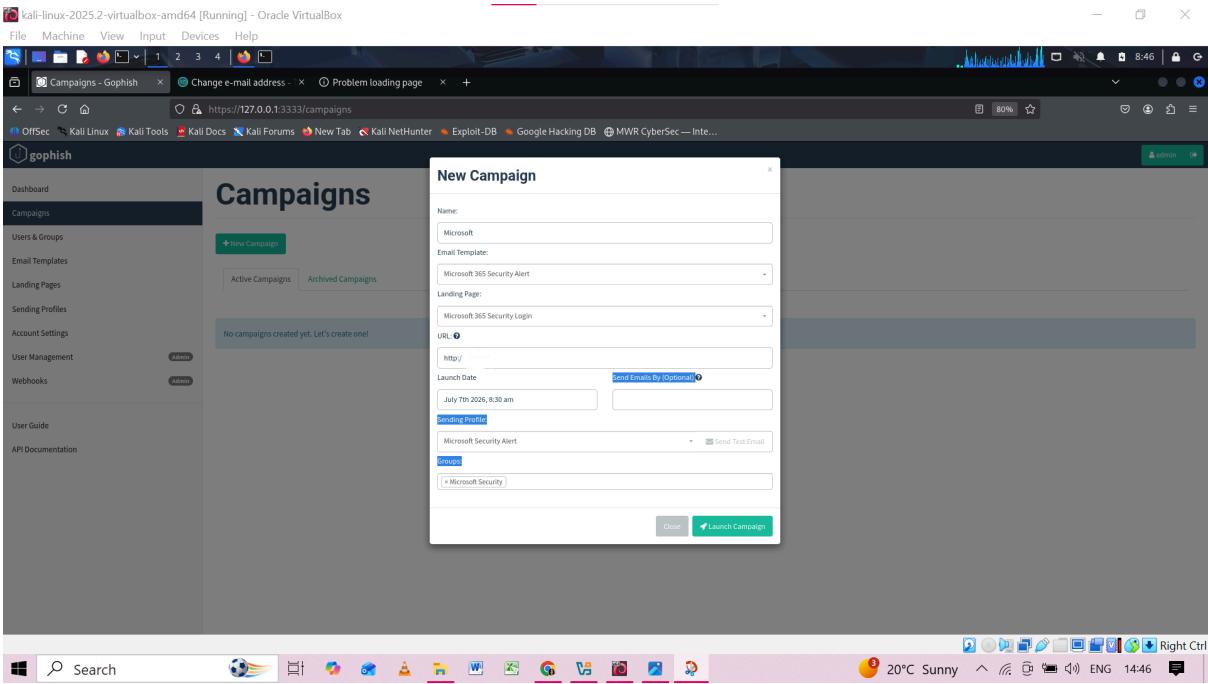
Phishing Port	80 (HTTP)
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Note on Development Process: As a beginner in web development, I leveraged AI tools to assist with HTML/CSS coding for the email templates and landing pages. This approach allowed me to focus on the security and infrastructure aspects of the project while ensuring the front-end components maintained a professional, realistic appearance. All AI-generated code was reviewed, tested, and customized to meet the specific requirements of the phishing simulation.

7. Campaign Execution

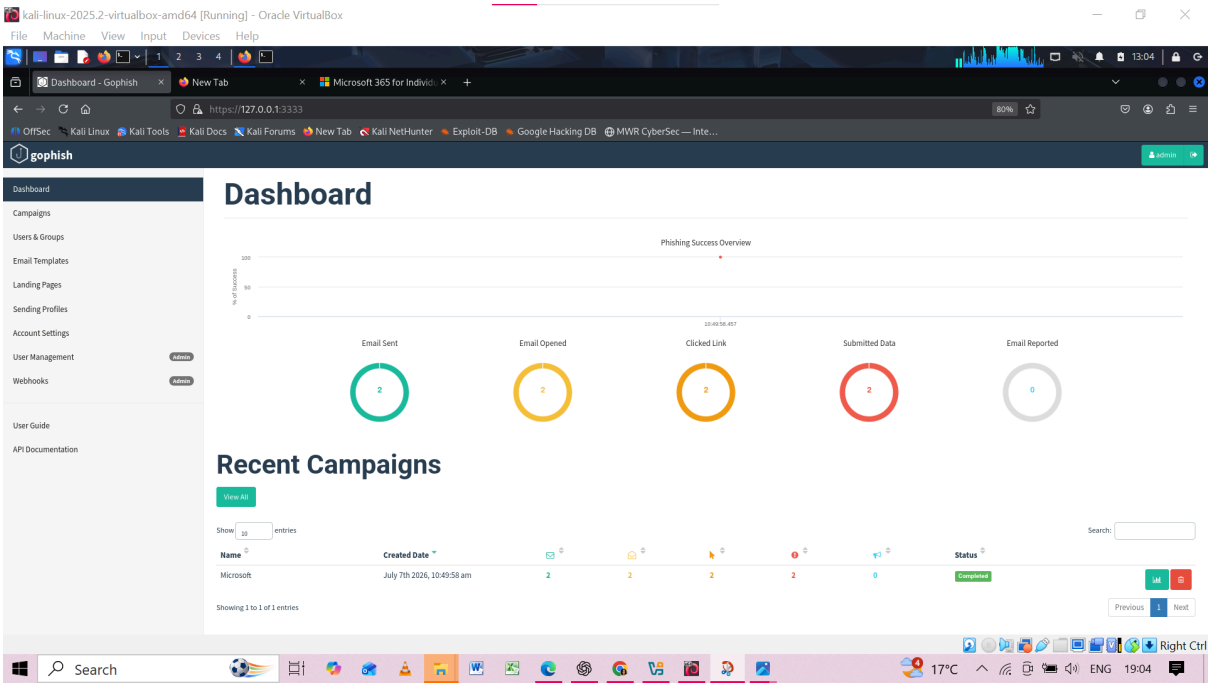
7.1 Campaign Setup

Figure 4: Campaign Configuration



7.2 Results Dashboard

Figure 5: Campaign Results Overview



7.3 Credential Capture

Figure 6: Captured Credentials

kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

Microsoft - Gophish

New Tab

Microsoft 365 for Individ...

https://127.0.0.1:3333/campaigns/4

80%

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gophish

Admin

Dashboard

Campaigns

Users & Groups

Email Templates

Landing Pages

Sending Profiles

Account Settings

User Management

Webhooks

User Guide

API Documentation

Details

Show 10 entries

Search:

First Name	Last Name	Email	Position	Status	Reported
John	Bendo	vi@v.com	Subscriber	Submitted Data	

Timeline for John Bendo

Email ID: vi@v.com

Result ID: vi@v.com

Campaign Created

July 7th 2026 10:49:58 am

Email Sent

July 7th 2026 10:49:59 am

Clicked Link

July 7th 2026 11:33:13 am

Submitted Data

July 7th 2026 11:33:32 am

Linux (OS Version: x86_64)

Firefox (Version: 128.0)

Replay Credentials

View Details

Parameter	Value(s)
email	vi@v.com
password	bhgh

Search

17°C

ENG

18:57

kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

Microsoft - Gophish

New Tab

Microsoft 365 for Individ...

https://127.0.0.1:3333/campaigns/4

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gophish

Admin

Dashboard

Campaigns

Users & Groups

Email Templates

Landing Pages

Sending Profiles

Account Settings

User Management

Webhooks

User Guide

API Documentation

Details

Show 10 entries

Search:

First Name	Last Name	Email	Position	Status	Reported
John	Bendo	vi@v.com	Subscriber	Submitted Data	
Nobuhle	Alex	teh@v.com	Premium Member	Submitted Data	

Timeline for Nobuhle Alex

Email ID: teh@v.com

Result ID: teh@v.com

Campaign Created

July 7th 2026 10:49:58 am

Email Sent

July 7th 2026 10:49:59 am

Clicked Link

July 7th 2026 11:16:13 am

Submitted Data

July 7th 2026 11:16:56 am

Linux (OS Version: x86_64)

Firefox (Version: 128.0)

Replay Credentials

View Details

Parameter	Value(s)
email	teh@v.com
password	hjsl

Search

17°C

ENG

18:59

8. Attack Flow Diagram

text

Email Sent (14:49:59)

Target Opens Email

↓

Clicks "Verify Your Account" Button (15:16:13)

↓

Landing Page Loads (Microsoft 365 Clone)

↓

Target Enters Credentials

↓

Credentials Captured by GoPhish (15:16:56)

↓

Redirected to Real Microsoft (www.microsoft.com)

9. Analysis & Findings

Key Findings:

1. 100% Click Rate: Both targets clicked the link immediately
2. 100% Submission Rate: Both targets entered credentials
3. Rapid Response: First submission within 25 minutes
4. No Verification: Targets did not verify the legitimacy

Why It Worked:

- Urgency: "Security Alert" created fear
- Familiarity: Microsoft branding looked legitimate
- Simple Action: One clear call-to-action button
- Convenience: Clicking was easier than investigating

Technical Observations:

- All targets used Firefox 128.0 on Linux
- Credentials were captured successfully
- Browser information was logged automatically

10. Recommendations

Short-Term:

1. Security Awareness Training
 - Regular phishing simulation training
 - How to identify red flags
 - Report suspicious emails
2. Multi-Factor Authentication (MFA)
 - Enable MFA for all accounts
 - Use authenticator apps not SMS

Long-Term:

1. Email Filtering
 - Advanced spam filters
 - Domain authentication (SPF, DKIM, DMARC)
2. Incident Response
 - Clear reporting process
 - Quick password resets
 - Account lockout procedures

11. Skills Demonstrated

Skill

How It Was Demonstrated

Social Engineering

Crafted realistic security alert email

Phishing Simulation

Used GoPhish to orchestrate campaign

Linux Administration

Kali Linux setup and configuration

Email Infrastructure

Configured Postfix SMTP server

Web Development

Created Microsoft 365 clone

Data Analysis

Analyzed campaign metrics

12. Conclusion

This project successfully demonstrated the effectiveness of phishing simulations in identifying security vulnerabilities. The 100% success rate highlights the critical need for comprehensive security awareness training and technical controls to protect against social engineering attacks.

13. Screenshots

Figure 1: Email Template Preview

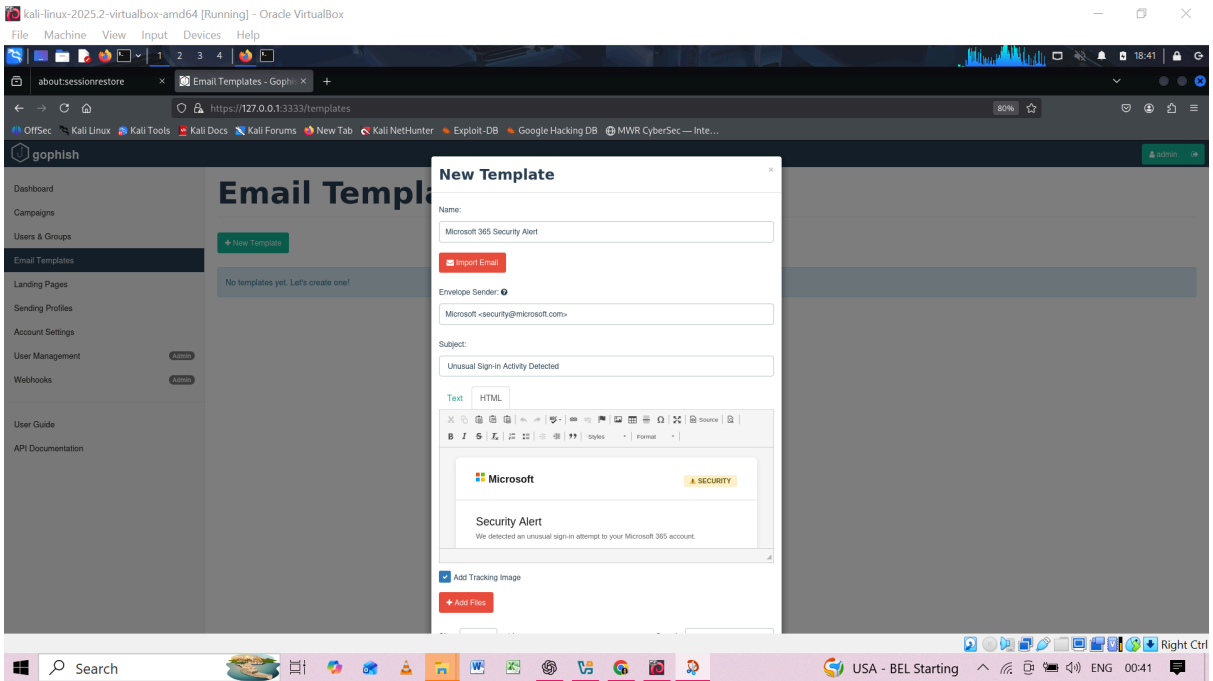
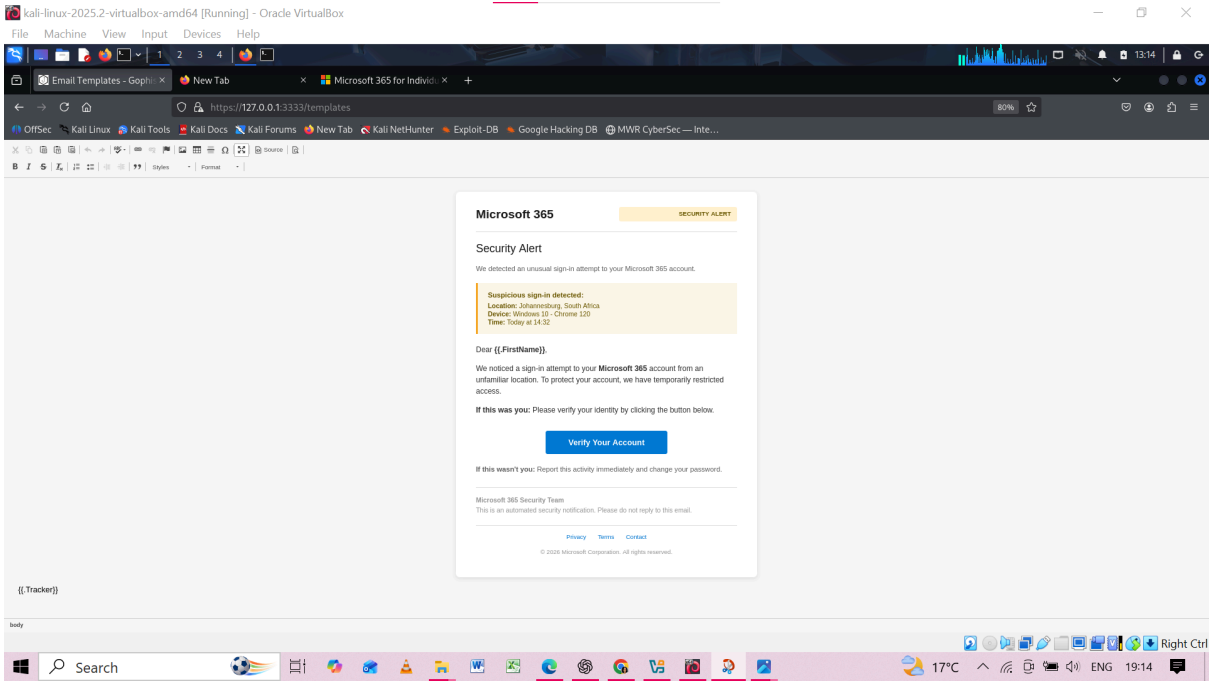


Figure 2: Landing Page

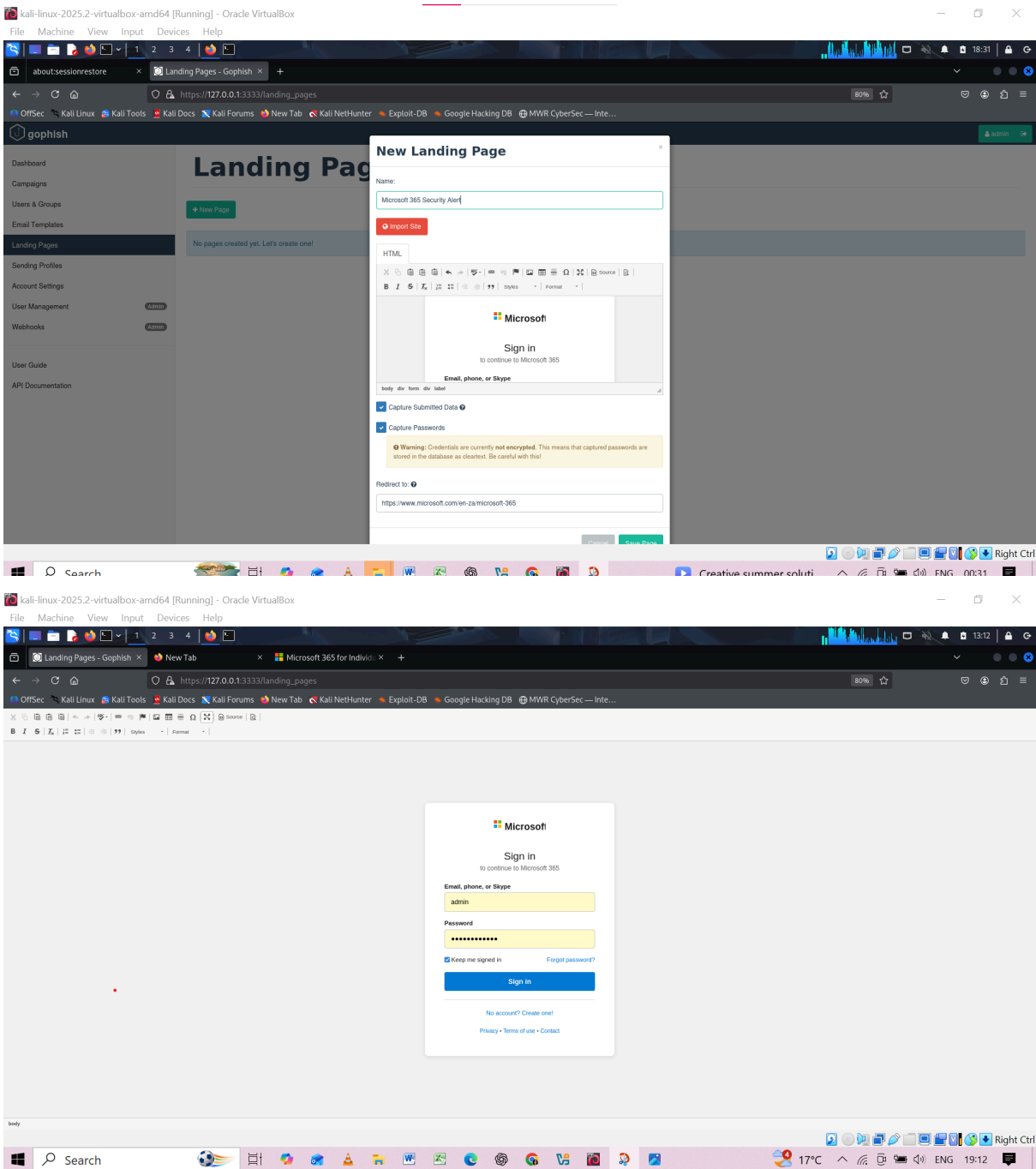


Figure 3: Sending Profile Configuration

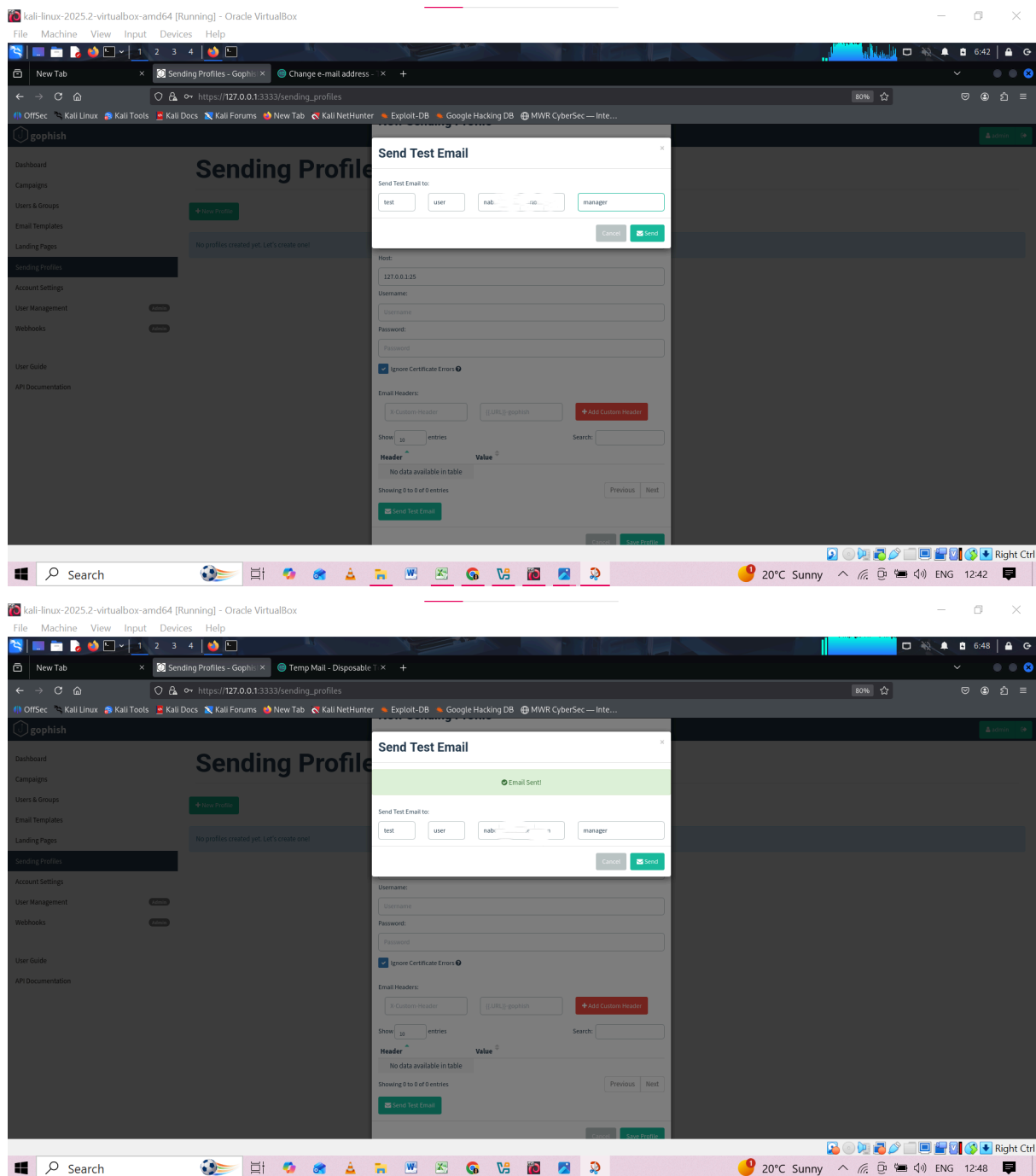


Figure 4: Campaign Configuration

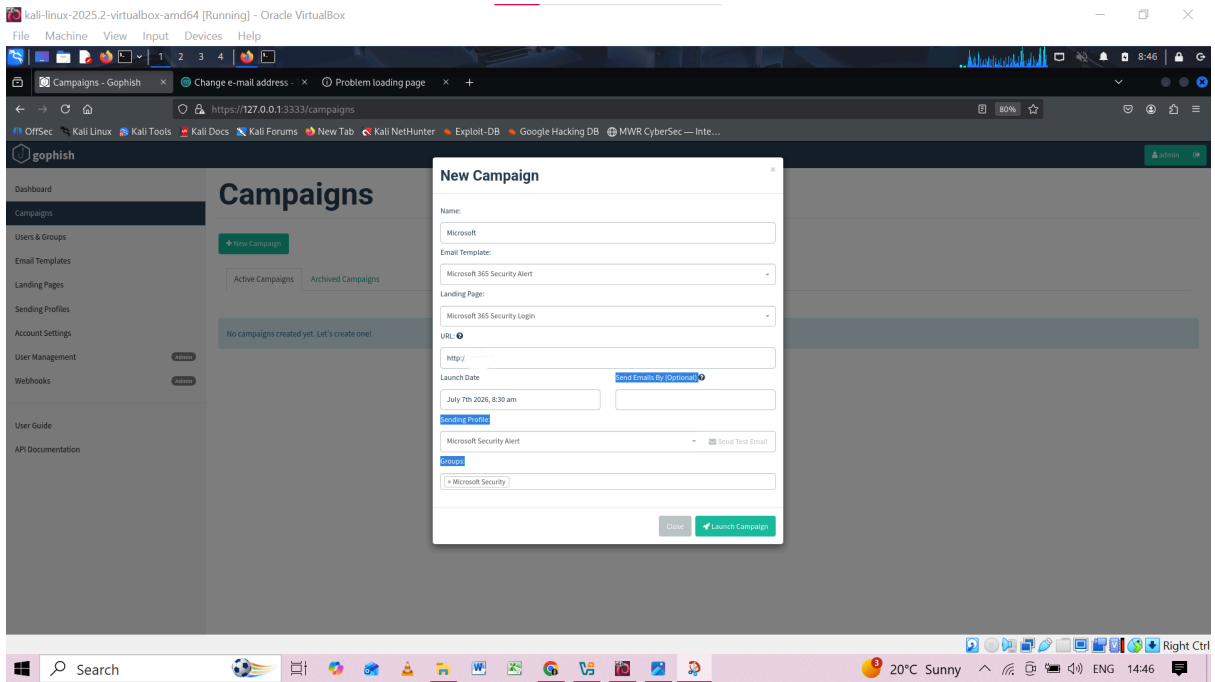


Figure 5: Campaign Results Overview

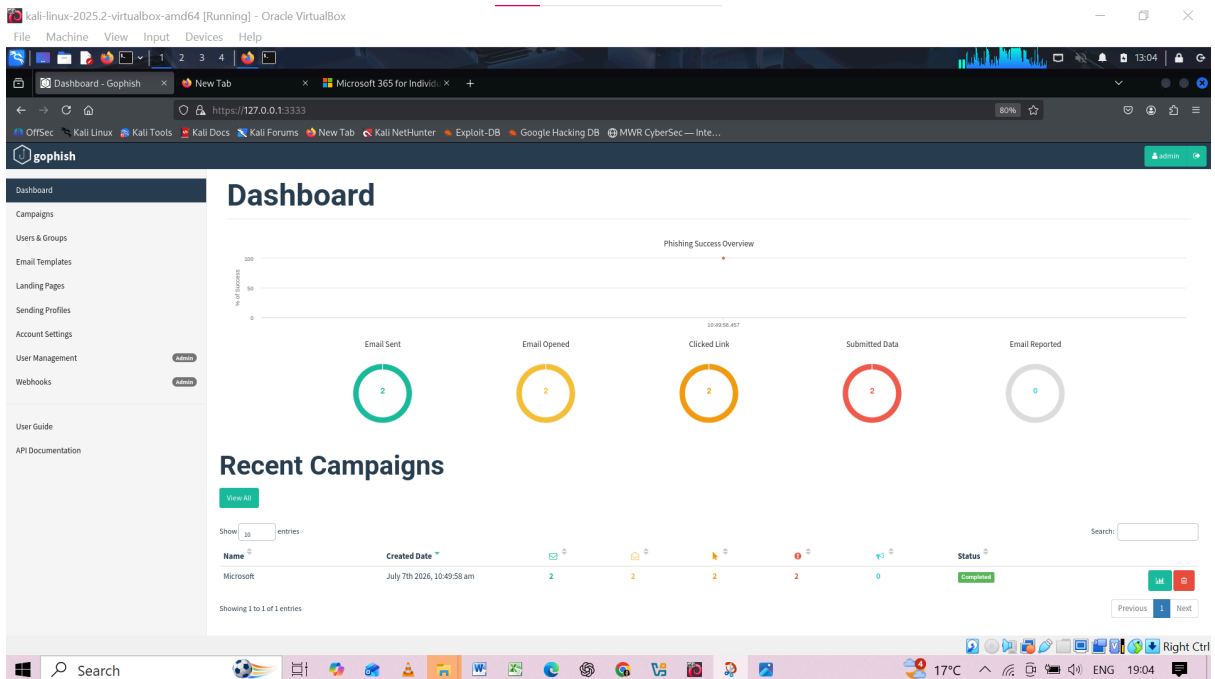


Figure 6: Captured Credentials

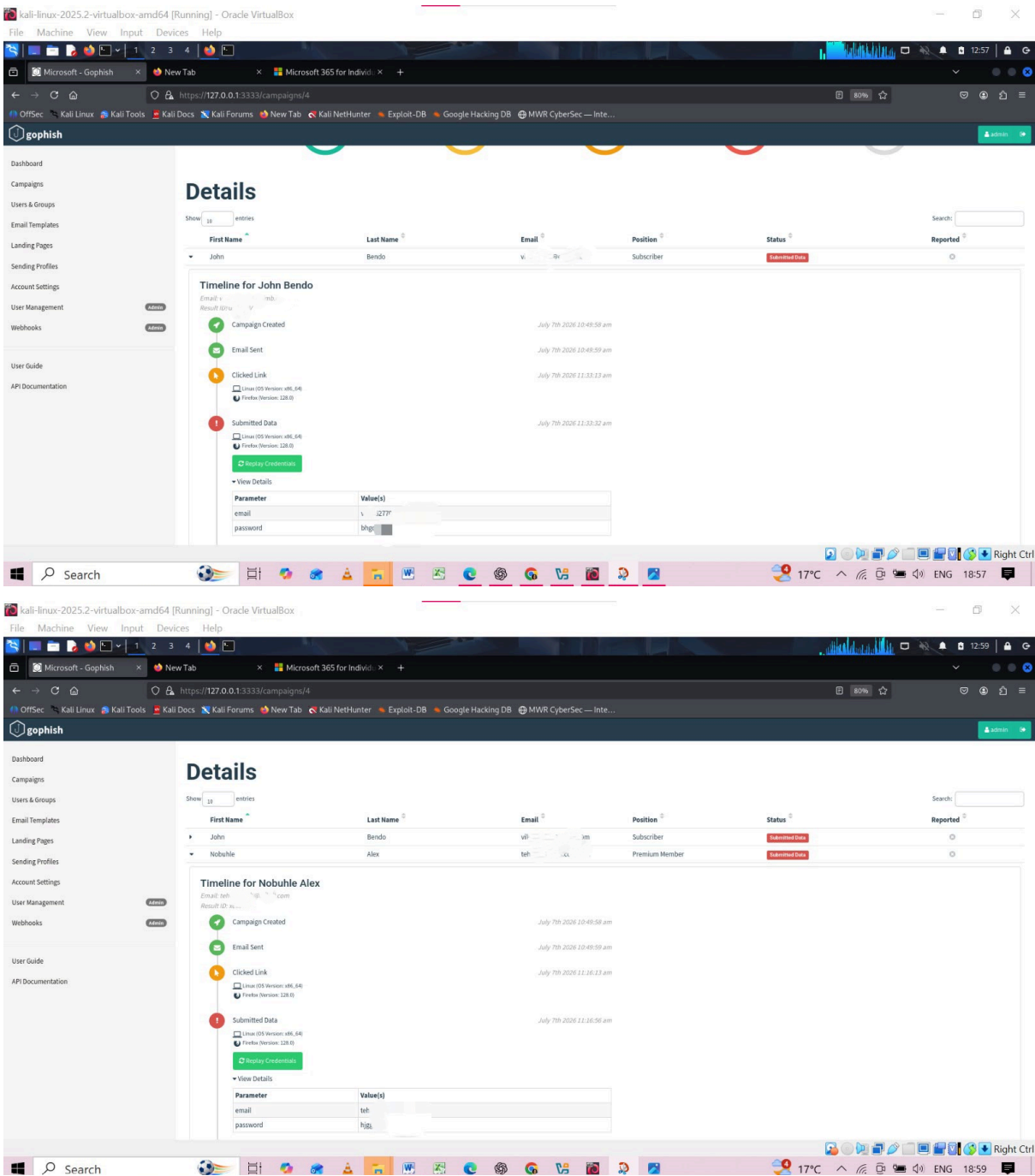


Figure 7: Temporary Email Inbox

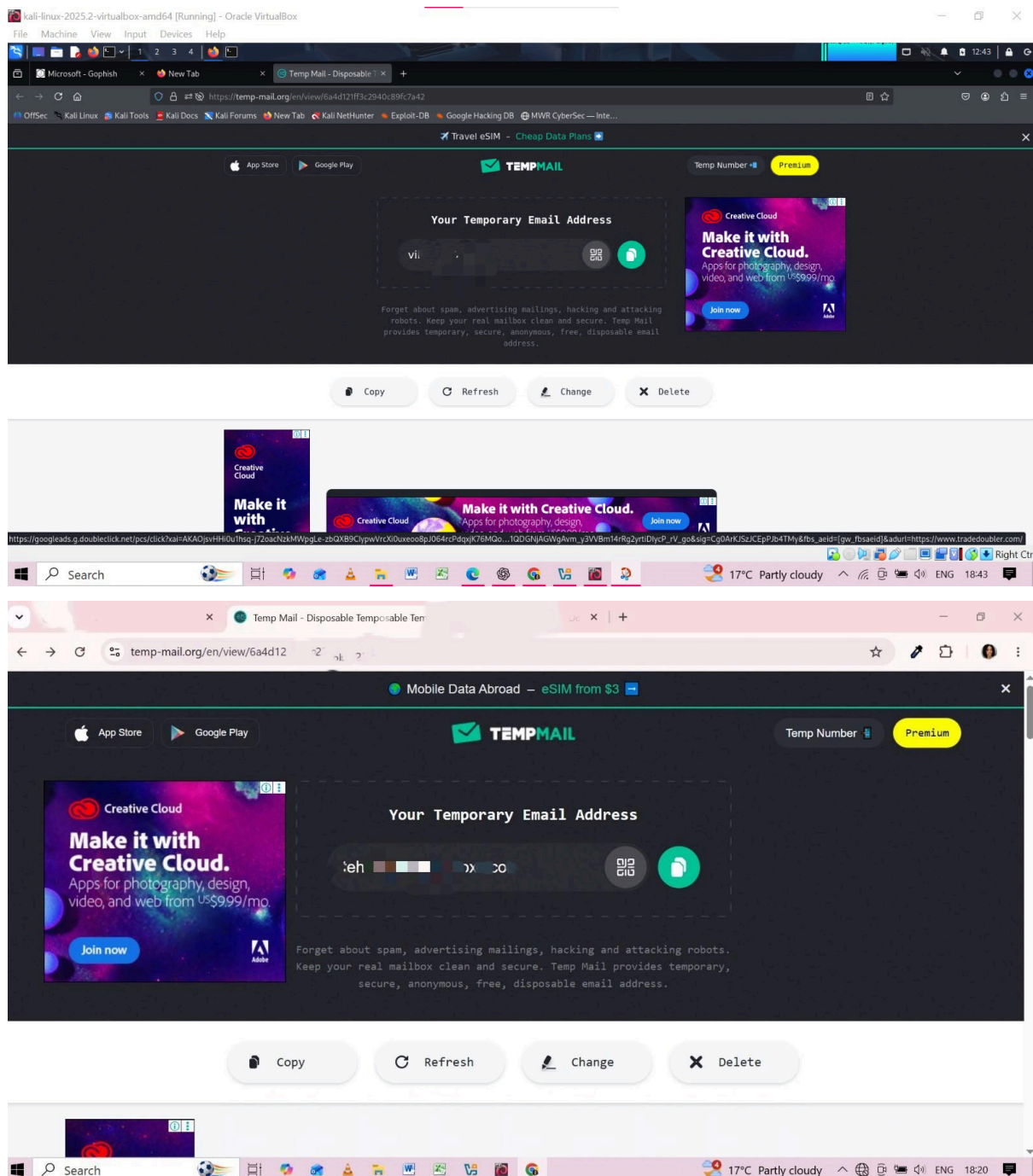


Figure 8: Users & Groups

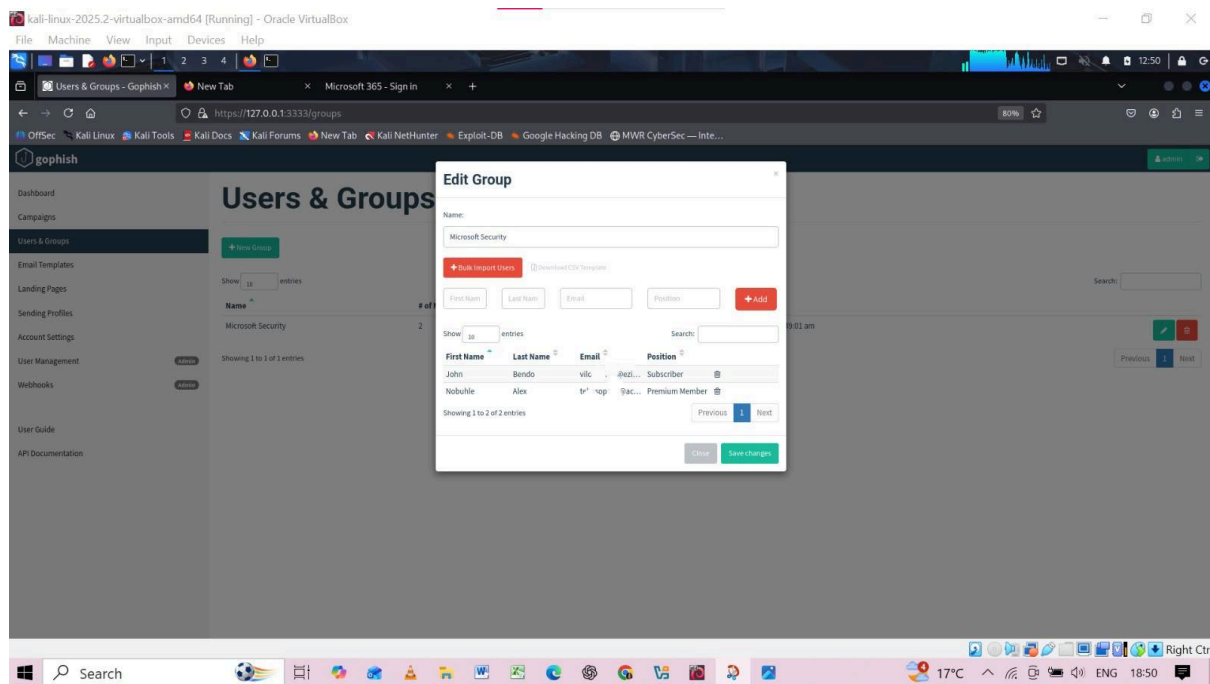


Figure 9: Postfix Installation

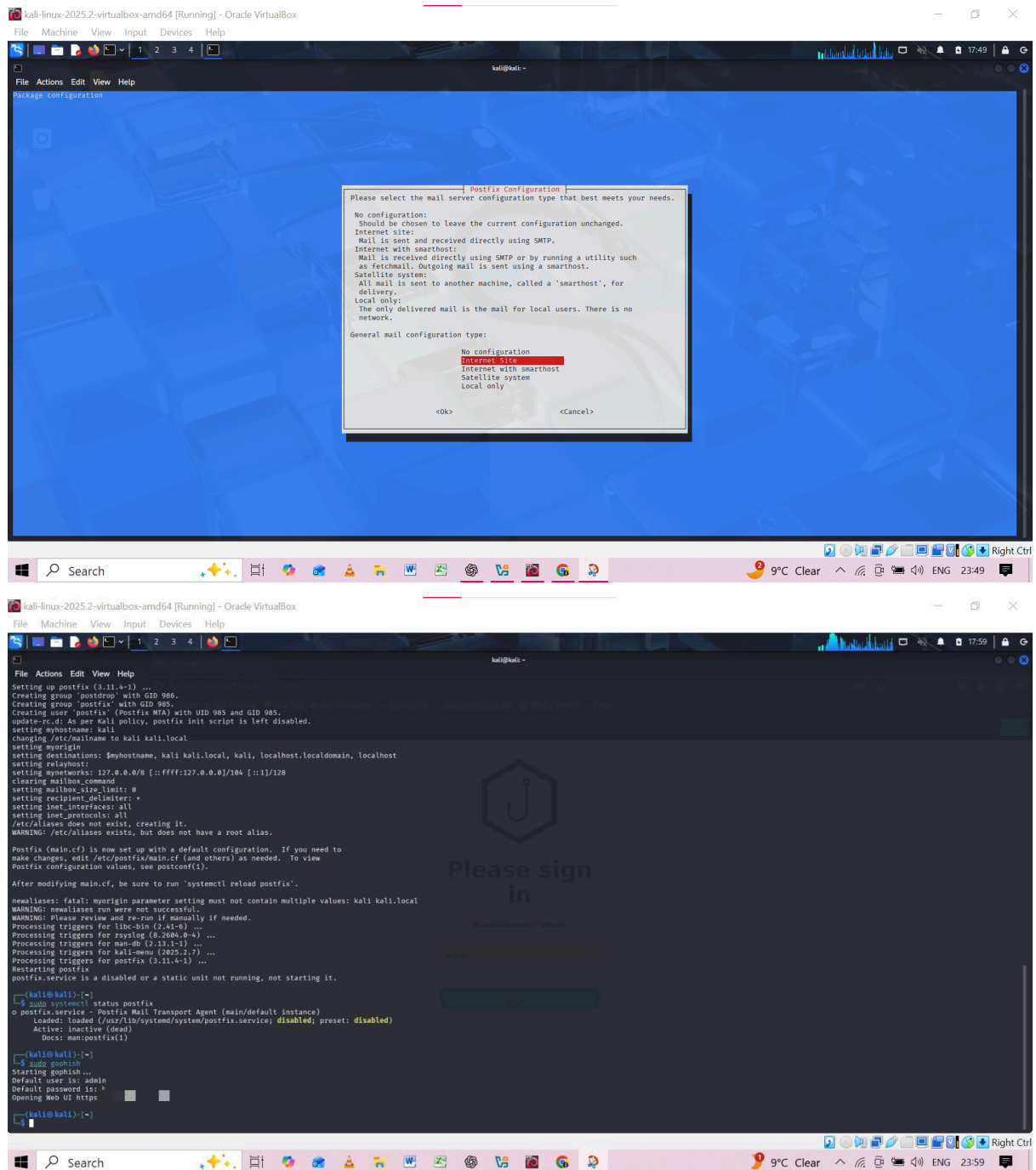


Figure 10: GoPhish Login

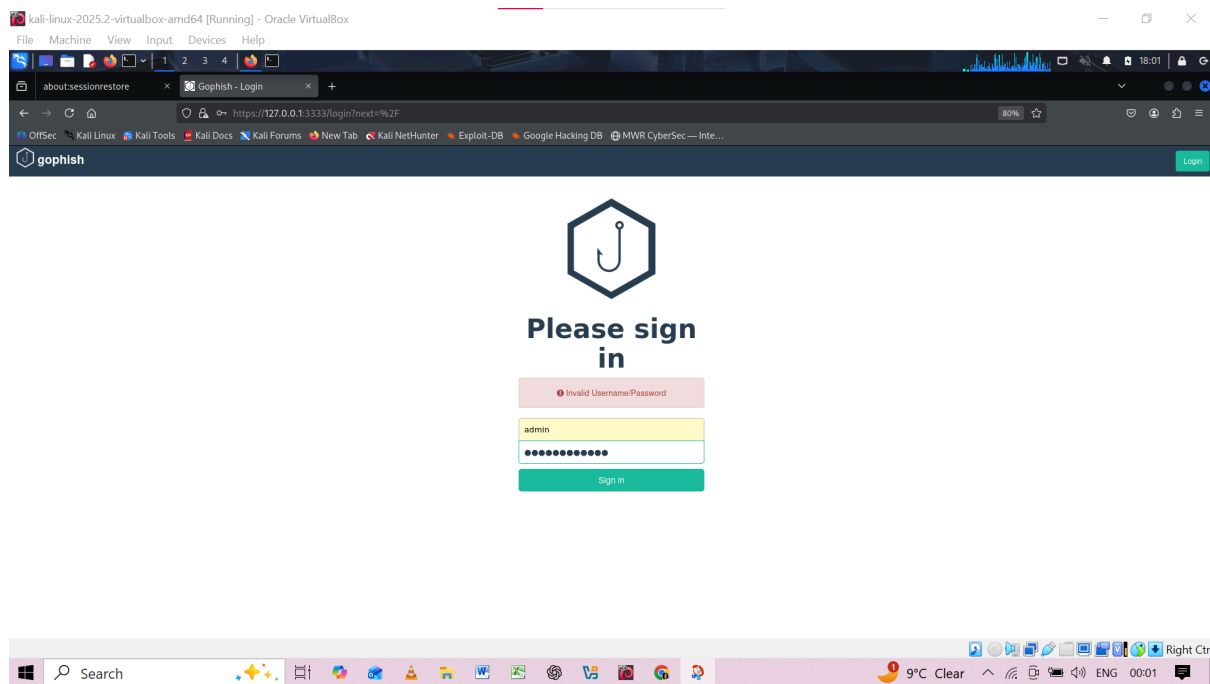
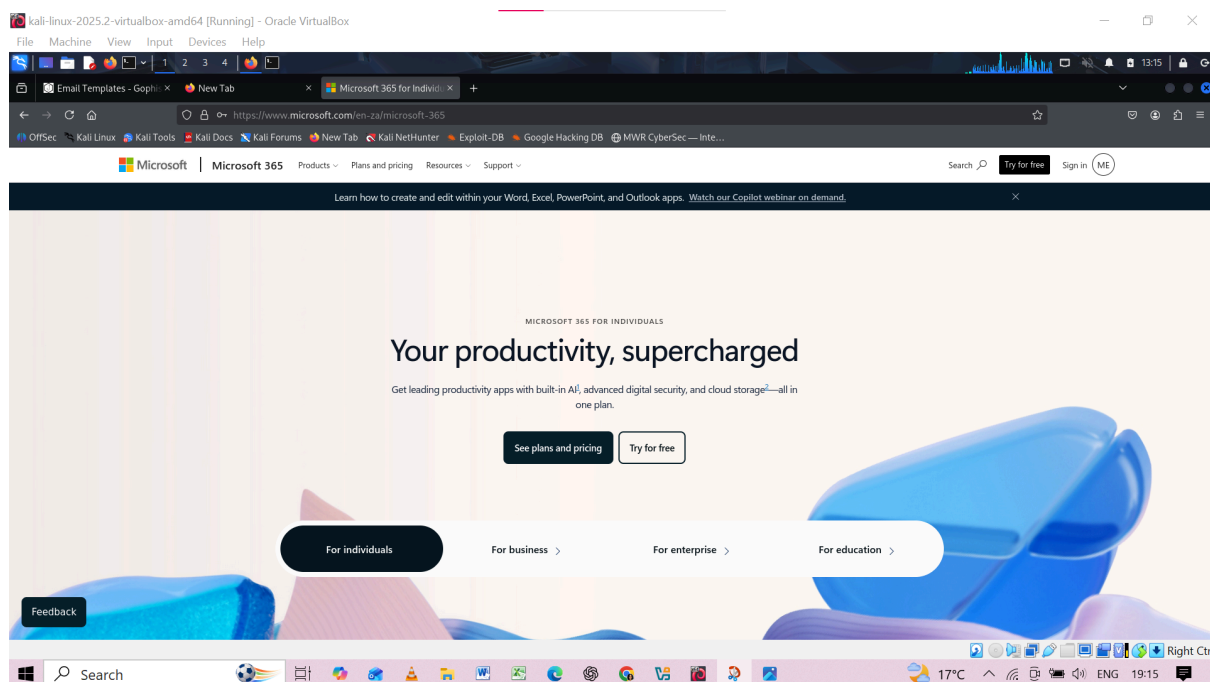


Figure 11: Redirect to real Microsoft website



Appendix B: Raw Campaign Data

campaign_id: 2

Total Emails Sent: 2

Total Opens: 2

Total Clicks: 2

Total Submissions: 2

Captured Credentials:

- t*****@a***s.*** / **gg*****

- v***6*****@e****.*** / b**d*****

Browser: Firefox 128.0 on Linux (x86_64)

Lab Server IP: 10.0.2.*.**

Report Generated: July 7, 2026

Prepared By: Galaletsang Mpye

Project: Phishing Simulation Campaign